

Question 4c

September 9, 2026

```
[1]: import scipy
import numpy as np
import matplotlib.pyplot as plt
```

```
[2]: def plot_u_and_report(I_0_crits):
    for I_0_crit, label in I_0_crits:
        def leaky_integrate_and_fire_model(t, u):
            dudt = (-u + I_0_crit*np.exp(-t/2))/10
            return dudt

        u0 = [0]
        t_span = (0, 50)
        t_eval = np.linspace(0, 50, 1000000)
        solution = scipy.integrate.solve_ivp(fun = leaky_integrate_and_fire_model, t_span = t_span, y0 = u0, t_eval = t_eval, rtol = 1e-10, atol = 1e-10)

        t_peak = solution.t[np.argmax(solution.y[0])]
        u_max = np.max(solution.y[0])

        print("-" * 50)
        print(f"I_0_crit : {I_0_crit}")
        print(f"t_peak: {t_peak}")
        print(f"u_max: {u_max}")
        print("-" * 50)

        plt.plot(solution.t, solution.y[0], label = label)

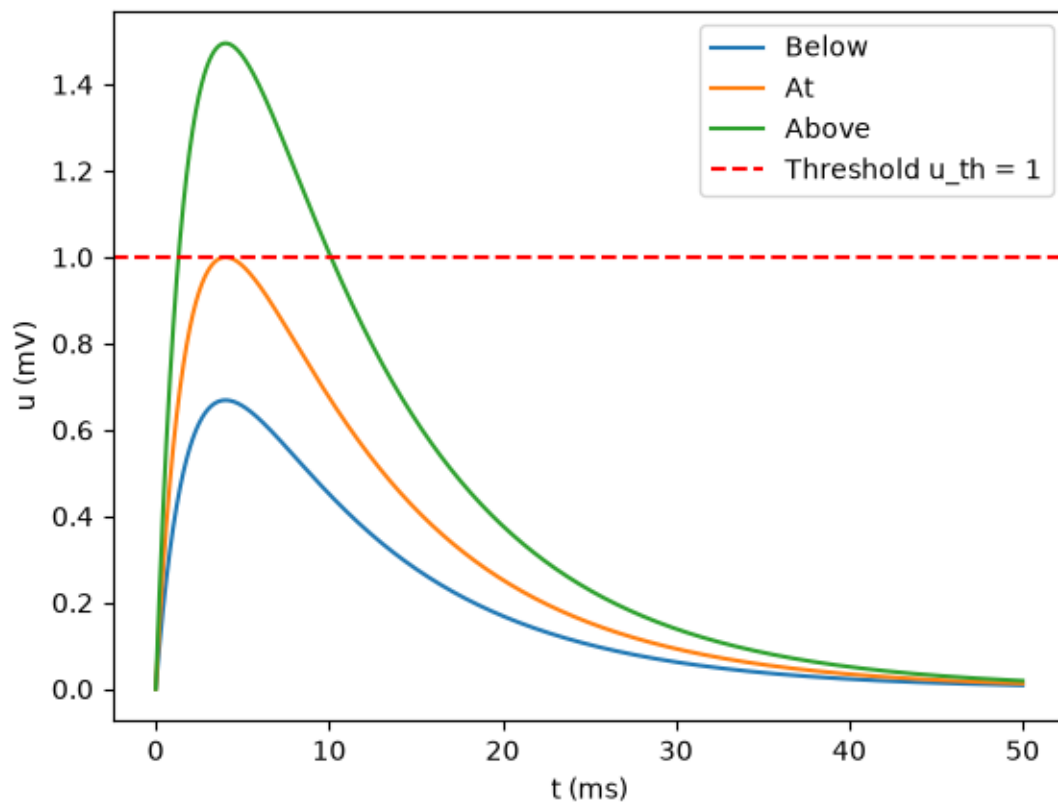
        plt.axhline(y = 1, color = "red", linestyle = "--", label = "Threshold u_th = 1")
        plt.xlabel("t (ms)")
        plt.ylabel("u (mV)")
        plt.legend()
        plt.show()
```

```
[3]: plot_u_and_report([(5, "Below"), (5**(5/4), "At"), (5**(6/4), "Above")])
```

```
I_0_crit : 5  
t_peak: 4.023604023604023  
u_max: 0.6687403046431972
```

```
-----  
I_0_crit : 7.476743906106103  
t_peak: 4.023604023604023  
u_max: 0.9999999995814137
```

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-----  
I_0_crit : 11.180339887498949  
t_peak: 4.023604023604023  
u_max: 1.4953487807574712
```



```
[4]: def plot_I_0_crit():  
      def calculate_I_0_crit(tau_s):  
          return (tau_s/10)**(1/(tau_s/10 - 1))  
      tau_s = np.linspace(0.5, 100, 100000)  
      I_0_crit = calculate_I_0_crit(tau_s)
```

```
plt.loglog(tau_s, I_0_crit)
plt.xlabel("$\\tau_s$ (ms)")
plt.ylabel("$I_0^{crit}$ $\\left(\\frac{\\mu A}{cm^2}\\right)$")
plt.show()
```

```
plot_I_0_crit()
```

